

Multimodal Time Series Models: A Survey and Outlook

Huirui Li

Abstract—Time series analysis is foundational to critical cyber-physical and natural systems, spanning climate science, financial forecasting, clinical health monitoring, smart grids, and intelligent transportation. While traditional paradigms and unimodal deep learning models rely exclusively on discrete numerical records, real-world systems are embedded in rich multimodal contexts—accompanied by news reports, natural language operational logs, visual charts, and spatial topologies. Recently, the confluence of Large Language Models (LLMs), Vision-Language Models (VLMs), and pre-trained Time Series Foundation Models (TSFMs) has catalyzed a paradigm shift toward Multimodal Time Series Models. This survey provides the first comprehensive, systematic review (following PRISMA 2020 guidelines) of multimodal time series research from 2021 to 2026. We formalize the multimodal time series problem, propose a multi-dimensional taxonomy decomposing the literature by modality pairing, fusion architecture, the functional role of complementary modalities, and downstream tasks. Furthermore, we categorize cutting-edge methodologies including cross-modal patch reprogramming, visual line-chart transcoding, unified multimodal tokenization, and neuro-symbolic reasoning agents. We benchmark the evolving dataset landscape, scrutinize persistent challenges—including cross-modal distribution gaps, temporal misalignment, and evaluation leakage—and outline high-leverage future research directions.

Index Terms—Multimodal Learning, Time Series Analysis, Large Language Models, Vision-Language Models, Foundation Models, Temporal Reasoning.



1 Introduction

Time series data represents the temporal pulse of our interconnected physical, biological, and socio-economic systems. Over the past several decades, the study of temporal sequence modeling has evolved across three major technological waves: classical statistical and autoregressive modeling (e.g., ARIMA, exponential smoothing), task-specific deep neural architectures (e.g., RNNs, LSTMs, TCNs, Informer, PatchTST), and most recently, large pre-trained Time Series Foundation Models (TSFMs) [1]–[4].

Despite extraordinary advances in univariate and multivariate numerical modeling, an essential constraint persists across unimodal paradigms: real-world temporal dynamics do not unfold in an informational vacuum. When a sudden traffic congestion spike occurs, an electrical grid frequency deviates, or a stock price undergoes rapid volatility, the underlying generative causes are rarely confined to historical numerical values alone. Instead, these phenomena are profoundly shaped by external multimodal signals: breaking news reports, meteorological alerts, natural language operational dispatches, visual monitoring streams, and relational graph topologies.

1.1 The Rise of Multimodal Time Series Intelligence

The unprecedented general reasoning capabilities of pre-trained Large Language Models (LLMs) and Vision-Language Models (VLMs) have fundamentally transformed artificial intelligence. Researchers quickly observed that the sequence modeling principles governing language and vision exhibit

deep mathematical parallels with time series. This has catalyzed an explosion of research exploring the intersection of time series and multimodal models:

- 1) **Cross-Modal Reprogramming:** Groundbreaking frameworks such as Time-LLM [5] and One Fits All (GPT4TS) [6] demonstrated that frozen language model backbones (e.g., GPT-2, LLaMA) can be repurposed as high-accuracy time-series forecasters by patching time series and aligning temporal representations with textual semantic prompts.
- 2) **Visual Transcoding & Vision-Language Synergy:** Counter-intuitively, VisionTS [7] revealed that reformulating numeric series as line-chart images allows off-the-shelf visual masked autoencoders (MAE pre-trained on ImageNet) to perform competitive zero-shot forecasting without any domain fine-tuning. Concurrently, Time-VLM [8] unites visual, textual, and temporal representations for augmented forecasting.
- 3) **Conversational Reasoning & TS-MLLMs:** Pioneering systems such as ChatTS [9], TimeOmni-1 [10], and ChatTime [11] elevate time-series models from passive function approximators to interactive reasoning agents capable of answering temporal questions, conducting counterfactual reasoning, and explaining anomalies.
- 4) **Multimodal Datasets & Benchmarks:** Systematic benchmarks such as Time-MMD [12], Fidel-TS [13], and MTBench [14] have begun curating high-fidelity, aligned cross-modal time series corpora across diverse real-world domains.

H. Li is with the School of Computer Science. Project repository and continuous review updates: <https://github.com/lihuirui/awesome-multimodal-time-series-survey>. Correspondence: 58367737+lihuirui@users.noreply.github.com.

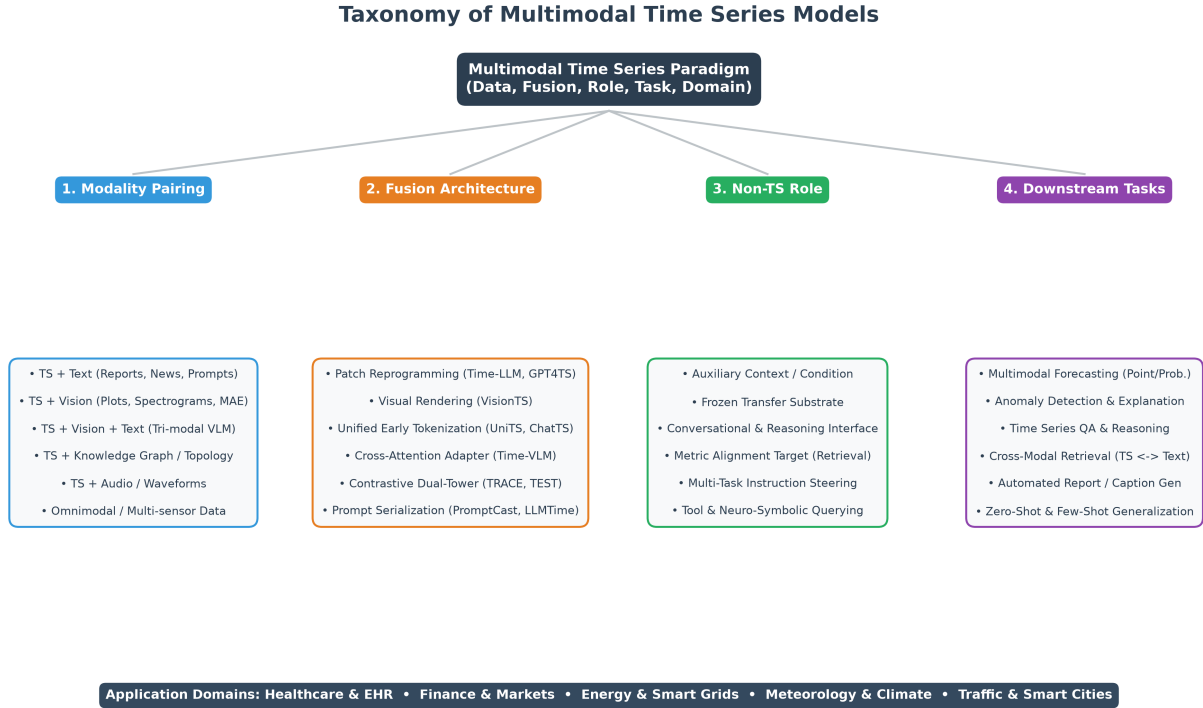


Fig. 1. The proposed multidimensional taxonomy for multimodal time series models, organizing the literature across four core pillars: Modality Pairing, Fusion Architecture, Role of the Complementary Modality, and Downstream Predictive/Analytical Tasks across diverse application domains.

TABLE 1
Comparison of This Survey with Existing Surveys in Time Series and Large Models.

Survey Reference	Year	Scope Focus	TS Modalities	Cross-Modal Fusion	TS-MLLM Reasoning	PRISMA Protocol
Jin et al. [15]	2023	Large Models (TS & ST)	TS + Spatio-Temporal	Partial	Limited	No
Jin et al. [16]	2024	Position: LLMs for TS	Numerical TS	Reprogramming focus	Preliminary	No
This Survey	2026	Multimodal TS Models	TS + Text + Vision + KG	Comprehensive	In-Depth (2021–2026)	Yes (PRISMA 2020)

1.2 Need for a Dedicated Survey

While prior comprehensive surveys have examined large foundation models for pure time-series or spatio-temporal data [15], [16], they predominantly focus on unimodal foundation architectures or single-modality adaptation. As shown in Table 1, there lacks a dedicated, systematic survey synthesizing the rapid evolution of explicitly multimodal time series modeling—spanning text-guided forecasting, vision-based time series modeling, multi-task unified architectures, and conversational reasoning agents.

1.3 Contributions of This Survey

To fill this critical void, this paper provides:

- A Systematic, Verifiable Review: Conducted under the PRISMA 2020 framework spanning 2021 to 2026, where every cited paper is verified against authoritative bibliographic APIs.
- A Four-Pillar Taxonomy: A taxonomy deconstructing multimodal time series research along Modality Pairing, Fusion Architecture, Non-TS Role, and Downstream Tasks (Figure 1).

- Deep Methodological Analysis: A unified comparative dissection of cross-modal reprogramming, visual line-chart transcoding, contrastive alignment, and interactive reasoning architectures.
- Curated Resource & Dataset Landscape: An exhaustive mapping of multimodal benchmark datasets, open-source repositories, and evaluation protocols.
- Critical Perspectives on Open Challenges: A rigorous appraisal of the modality gap, benchmark data contamination, temporal granularity mismatches, and future research trajectories.

2 Background & Preliminaries

In this section, we formally formulate the multimodal time-series modeling problem, detail the standard mathematical representations of temporal and non-temporal modalities, and review foundational learning objectives.

2.1 Problem Formulation

Definition 1 (Multivariate Time Series). Let $\mathbf{X}_{1:T} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_T]^T \in \mathbb{R}^{T \times C}$ denote a continuous or dis-

crete multivariate temporal sequence observed over T uniform time steps across C feature channels, where $\mathbf{x}_t = [x_{t,1}, x_{t,2}, \dots, x_{t,C}]^\top \in \mathbb{R}^C$.

Definition 2 (Complementary Multimodal Context). In a multimodal setting, the time series $\mathbf{X}_{1:T}$ is paired with complementary non-temporal contextual information $\mathcal{M} = \{\mathbf{M}_{\text{text}}, \mathbf{M}_{\text{vis}}, \mathbf{M}_{\text{graph}}\}$, where:

- $\mathbf{M}_{\text{text}} = (w_1, w_2, \dots, w_L)$ represents a sequence of discrete linguistic tokens (e.g., domain descriptions, incident logs, financial news, or analytical instructions).
- $\mathbf{M}_{\text{vis}} \in \mathbb{R}^{H \times W \times 3}$ denotes a 2D visual representation (e.g., plotted line charts, spectrogram transforms, or synchronous satellite imagery).
- $\mathbf{M}_{\text{graph}} = (\mathcal{V}, \mathcal{E}, \mathbf{A})$ encapsulates topological or spatial relationships connecting multivariate time-series nodes.

Definition 3 (Multimodal Time Series Objective). Given an observed history $\mathbf{X}_{1:T}$ and context $\mathcal{M}$, the goal is to parameterize a model f_Θ that maps $\{\mathbf{X}_{1:T}, \mathcal{M}\}$ to a target space $\mathbf{Y}$ (which may represent future horizons $\mathbf{X}_{T+1:T+H}$, anomaly labels $\mathbf{y} \in \{0, 1\}^T$, textual answers, or joint embeddings) by minimizing an expected loss:

$$\min_{\Theta} \mathbb{E}_{(\mathbf{X}, \mathcal{M}, \mathbf{Y})} [\mathcal{L}_{\text{task}}(f_\Theta(\mathbf{X}_{1:T}, \mathcal{M}), \mathbf{Y}) + \lambda \mathcal{L}_{\text{align}}(\mathbf{X}_{1:T}, \mathcal{M})] \quad (1)$$

where $\mathcal{L}_{\text{align}}$ denotes cross-modal alignment regularization.

2.2 Temporal Patching and Tokenization

Direct point-wise numerical tokenization often suffers from high computational overhead and lacks semantic density. Following modern practice [5], [6], continuous time series are transformed into N temporal patches of length P with stride S :

$$\mathbf{P}_i = \mathbf{x}_{(i-1)S:(i-1)S+P} \in \mathbb{R}^{P \times C}, \quad i \in \left\{1, \dots, N = \left\lfloor \frac{T-P}{S} \right\rfloor + 1\right\} \quad (2)$$

Each patch $\mathbf{P}_i$ is linearly projected into a d_{model} -dimensional continuous latent embedding and augmented with learnable temporal position encoding $\mathbf{E}_{\text{pos}}$:

$$\mathbf{h}_i = \mathbf{P}_i \mathbf{W}_{\text{in}} + \mathbf{E}_{\text{pos},i}, \quad \mathbf{W}_{\text{in}} \in \mathbb{R}^{(P \cdot C) \times d_{\text{model}}} \quad (3)$$

2.3 Cross-Modal Alignment Paradigms

Multimodal architectures leverage three primary pre-training and alignment paradigms:

- 1) **Contrastive Representation Learning (InfoNCE):** Given paired time series representations $\mathbf{z}^{\text{ts}}$ and context representations $\mathbf{z}^{\text{text}}$, a dual-encoder minimizes the normalized temperature-scaled cross-entropy [17], [18]:

$$\mathcal{L}_{\text{InfoNCE}} = -\log \frac{\exp(\text{sim}(\mathbf{z}_i^{\text{ts}}, \mathbf{z}_i^{\text{text}})/\tau)}{\sum_{j=1}^B \exp(\text{sim}(\mathbf{z}_i^{\text{ts}}, \mathbf{z}_j^{\text{text}})/\tau)} \quad (4)$$

- 2) **Masked Autoencoding (MAE):** In visual or temporal reconstruction, a subset of patches $\mathcal{M}_{\text{mask}} \subset \{1, \dots, N\}$ is masked out, and the model reconstructs the unobserved temporal values from visible tokens [7]:

$$\mathcal{L}_{\text{MAE}} = \frac{1}{|\mathcal{M}_{\text{mask}}|} \sum_{i \in \mathcal{M}_{\text{mask}}} \|\mathbf{P}_i - \hat{\mathbf{P}}_i\|_2^2 \quad (5)$$

- 3) **Autoregressive Next-Token Prediction:** In decoder-only LLM-adapted forecasters, predictions are generated sequentially conditioned on prefix prompt tokens $\mathbf{E}_{\text{prompt}}$ and temporal patch tokens [9], [19]:

$$\mathcal{L}_{\text{AR}} = -\sum_{t=1}^K \log P(y_t | y_{<t}, \mathbf{E}_{\text{prompt}}, \mathbf{h}_{1:N}) \quad (6)$$

3 Taxonomy of Multimodal Time Series Models

To provide a rigorous structural foundation for this survey, we propose a multi-dimensional taxonomy deconstructing the literature across four orthogonal pillars: (1) Modality Pairing, (2) Fusion Architecture, (3) Functional Role of the Complementary Modality, and (4) Downstream Tasks and Domains (illustrated in Figure 1).

3.1 Pillar 1: Modality Pairing

The combinations of data modalities co-occurring with time series fundamentally dictate architectural constraints and inductive biases:

- **Time Series + Text (TS + Text):** The most prominent paradigm. Text serves diverse functions: semantic descriptions (e.g., sensor metadata, sampling rates, geographical origins in Time-LLM [5] and TEMPO [20]), continuous narrative reports (financial news, corporate filings, EHR medical notes in Time-MMD [12]), or natural language analytical queries [9], [21].
- **Time Series + Vision (TS + Vision):** Emerged prominently with VisionTS [7]. This pairing exploits 2D visual representations—either by directly plotting numerical trajectories as line charts or converting temporal signals into 2D time-frequency spectrograms—to harness vision foundation models pre-trained on massive natural image corpora.
- **Tri-Modal Systems (TS + Vision + Text):** Recent advanced architectures, notably Time-VLM [8], combine temporal historical records with synchronous visual observations (e.g., traffic camera feeds, satellite imagery) and textual context to form holistic multi-sensory representations.
- **Time Series + Spatial Topologies / Knowledge Graphs:** Integrating graph adjacency structures with sensor series to constrain spatio-temporal predictions through physical laws or topological connectivity [15].

3.2 Pillar 2: Fusion Architecture

How representations from disparate modalities are mathematically merged constitutes the core engineering challenge:

- 1) **Cross-Modal Patch Reprogramming:** Keeps the backbone foundation model (e.g., LLaMA, GPT-2) completely frozen. Patch-level linear or multi-head attention projection layers reprogram numerical time-series patches into the continuous embedding space of language tokens [5], [6], [22], [23].
- 2) **Modality Transcoding & Visual Rendering:** Transforms the temporal sequence into another modality prior to model ingestion. VisionTS [7] renders normalized time series into pixel bitmaps, completely

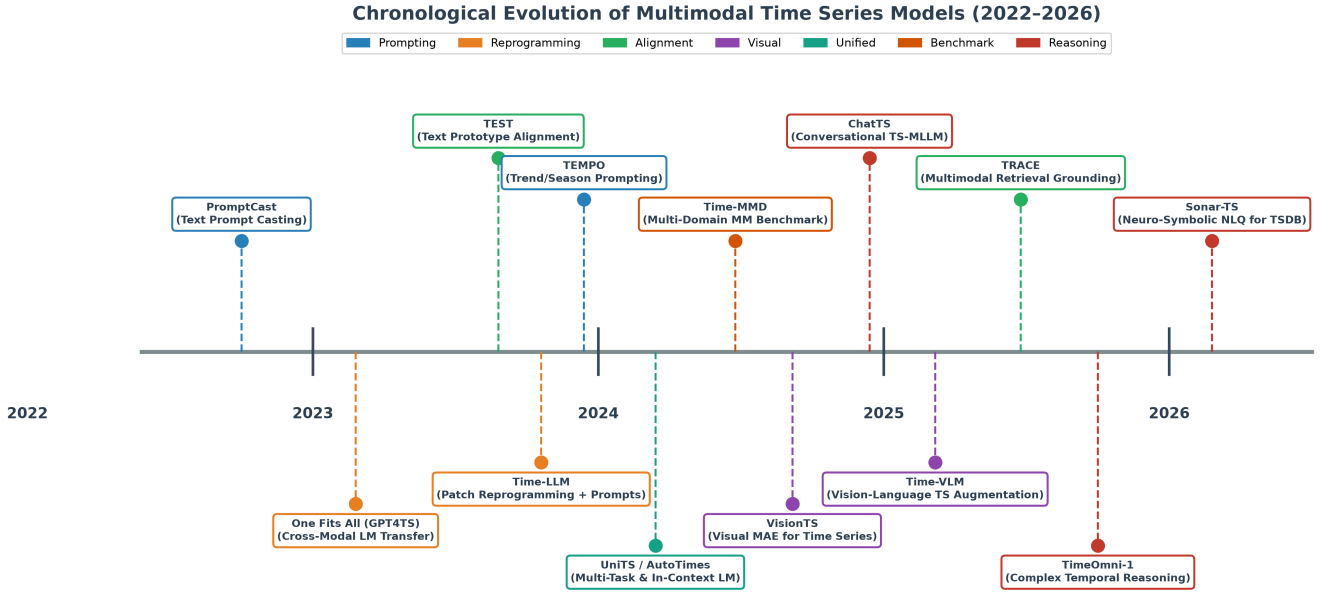


Fig. 2. Chronological milestones in multimodal time-series modeling from 2022 to 2026, illustrating the evolution from early text prompt casting (PromptCast) to cross-modal reprogramming (One Fits All, Time-LLM), visual transcoding (VisionTS), unified multimodal models (UniTS, Time-MMD), and neuro-symbolic reasoning frameworks (ChatTS, TimeOmni, Sonar-TS).

bypassing numerical tokenization and directly processing the visual stream with ViT-based Masked Autoencoders.

- 3) Unified Early Tokenization: Maps heterogeneous modalities into a shared token dictionary. Models such as UniTS [24] and ChatTime [11] tokenize time-series patches, instruction prompts, and domain markers into unified token embeddings passed jointly into deep Transformer stacks.
- 4) Cross-Attention & Adapter Modules: Employs dedicated adapter blocks where queries from one modality attend to keys and values from another modality, as exemplified by Time-VLM's dual-augmented learners [8].
- 5) Dual-Tower Metric Alignment: Maintains separate, dedicated encoders for time-series and textual context, aligning them in a shared latent metric space using contrastive losses (e.g., TRACE [18] and TEST [17]).

3.3 Pillar 3: Functional Role of the Complementary Modality

Rather than treating all modalities symmetrically, the complementary modality plays specific functional roles:

- Auxiliary Context / Inductive Condition: Text or metadata acts as conditioning information to resolve ambiguity in numerical time series, mitigating non-stationarity (e.g., Time-LLM, TEMPO, S²IP-LLM [25]).
- Frozen Transfer Substrate: The language model acts as an overparameterized pattern-matching backbone whose self-attention mechanisms transfer directly to universal time-series forecasting (One Fits All [6]).

- Conversational & Reasoning Interface: The non-TS modality serves as an interactive communication interface, enabling open-ended question answering, root-cause diagnostics, and counterfactual simulation (ChatTS [9], TimeOmni-1 [10]).
- Retrieval Anchor: Text serves as a semantic query anchor for indexing, clustering, and retrieving dynamic time-series trajectories (TRACE [18]).

3.4 Pillar 4: Downstream Tasks and Domains

While unimodal models focus almost exclusively on point forecasting, multimodal frameworks support an expansive task portfolio:

- Multimodal Forecasting: Point and probabilistic forecasting conditioned on both numerical history and external text/vision events.
- Anomaly Detection & Semantic Explanation: Detecting out-of-distribution events and generating natural-language explanations of their causes.
- Time-Series Question Answering (TS-QA): Answering queries regarding periodic trends, maximum peaks, and correlated shifts.
- Cross-Modal Retrieval: Retrieving relevant time-series segments given natural language descriptions, and vice versa.

These tasks span diverse sectors: Meteorology (Weather-Bench), Finance (stock volatility, earnings calls), Healthcare (EHR signals, ICU monitoring), Smart Grids (load and renewable generation), and Transportation (traffic sensor dynamics).

4 Methodological Landscape: Core Paradigms

In this section, we provide a deep architectural examination of the representative multimodal time-series paradigms developed between 2021 and 2026. A comprehensive comparison is presented in Table 2.

4.1 Cross-Modal Patch Reprogramming Frameworks

The foundational breakthrough in adapting pre-trained NLP backbones to time series without catastrophic forgetting was achieved via cross-modal reprogramming:

- **One Fits All (GPT4TS)** [6]: Zhou et al. demonstrated that by freezing the multi-head self-attention and feed-forward layers of a pre-trained GPT-2 backbone while tuning only the linear input projection, layer normalization parameters, and an output head, the model exhibits remarkable universal pattern-matching capabilities across forecasting, classification, anomaly detection, and imputation.
- **Time-LLM** [5]: Jin et al. extended this paradigm by recognizing that numerical time series inherently lack semantic context. Time-LLM introduces: (1) Prompt-as-Prefix, which encodes dataset-specific background knowledge, task specifications, and statistical summary statistics (trends, lags) into textual prompts; and (2) a multi-head cross-attention Patch Reprogramming Layer that projects input patches into the text embedding dictionary.
- **AutoTimes** [19]: Overcomes the fixed-horizon limitation of previous forecasters by casting time series as an in-context autoregressive generation task, conditioning on textual chronological timestamps to capture long-range calendar periodicities.

4.2 Modality Transcoding: The Visual Rendering Paradigm

A radical departure from language-based adaptation is the Visual Transcoding framework pioneered by VisionTS [7]. Rather than struggling to map continuous 1D temporal trajectories into 1D text tokenizers, VisionTS reformulates forecasting as an image reconstruction task:

- 1) The normalized historical lookback window and future horizon are rendered onto a 2D canvas as a continuous line chart.
- 2) A pre-trained Visual Masked Autoencoder (MAE) [7] treats the historical curve as visible image patches and the future forecast segment as masked patches.
- 3) Strikingly, because natural image pre-training on ImageNet instills strong priors over spatial continuity, edge tracing, and geometric smoothness, VisionTS achieves competitive zero-shot forecasting performance without requiring any domain-specific time series fine-tuning.

Expanding on visual synthesis, Time-VLM [8] combines visual time-frequency spectrogram representations with textual descriptions and raw temporal sequences, demonstrating that visual rendering and textual guidance provide complementary inductive biases.

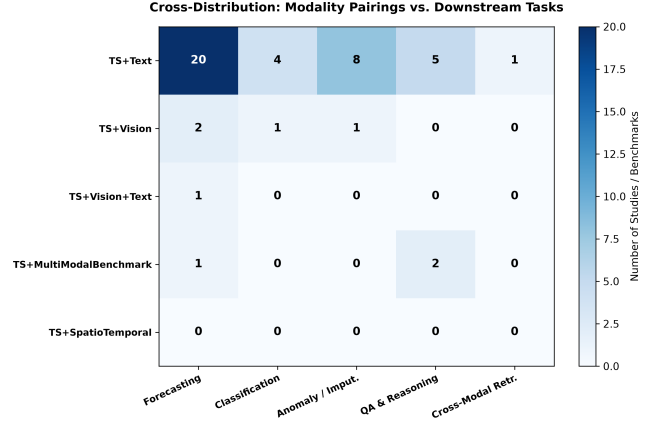


Fig. 3. Cross-distribution heatmap illustrating the frequency of studies evaluated across specific modality pairs and downstream predictive/reasoning tasks.

4.3 Conversational Multimodal LLMs & Reasoning Suites

Moving beyond classical regression metrics, recent research aims to empower foundation models with temporal reasoning capabilities:

- **ChatTS** [9]: Identifies that text serialization loses fine-grained numerical resolution while naive reprogramming lacks conversational flexibility. ChatTS implements native numerical tokenization coupled with a synthetic Time Series Evol-Instruct curriculum, enabling multi-turn dialogues, peak detection, and causal hypothesis formulation.
- **TimeOmni-1** [10]: Formulates the Time Series Reasoning Suite (TSR-Suite), structuring temporal intelligence across three hierarchical stages: Perception (identifying seasonality, trends, and change points), Extrapolation (forecasting and scenario generation), and Decision-Making (optimizing interventions based on temporal dynamics).

5 Datasets, Benchmarks, and Evaluation

Rigorous evaluation of multimodal time series models requires benchmarks that provide synchronized, high-quality pairings across modalities. In this section, we examine the dataset landscape, summarize standard evaluation protocols, and discuss cross-task distributions.

5.1 Benchmark Suites and Multimodal Corpora

As illustrated in Figure 4, early multimodal experiments relied on ad-hoc combinations of univariate benchmark series (e.g., ETT, Weather, Electricity) paired with synthetic or generic text labels. Recently, comprehensive multi-domain suites have emerged:

- **Time-MMD** [12]: Spanning 9 diverse domains (including finance, energy, weather, healthcare, traffic, and agriculture), Time-MMD provides fine-grained, timestamp-aligned textual context (e.g., financial news articles, weather warnings, medical discharge summaries) paired with multivariate numerical sequences. Empirical evaluations on Time-MMD demonstrate

TABLE 2
Comprehensive Comparison of Representative Multimodal Time Series Models.

Model	Venue / Year	Modality	Backbone Architecture	Fusion Mechanism	Pre-training / Loss	Zero-Shot	Open Source
PromptCast [21]	IEEE TKDE '23	TS + Text	T5 / BART	Text Serialization	Cross-Entropy (AR)	No	✓
One Fits All [6]	NeurIPS '23	TS + Text	Frozen GPT-2	Patch Reprogramming	MSE Loss	✓	✓
TEST [17]	arXiv '23	TS + Text	BERT / GPT-2	Contrastive Prototype	Soft InfoNCE	✓	✓
Time-LLM [5]	ICLR '24	TS + Text	Frozen LLaMA-7B / GPT-2	Reprogramming + Prompts	Normalized MSE	✓	✓
TEMPO [20]	ICLR '24	TS + Text	GPT-2	Decomposed Prompting	MSE + Prompt Tuning	✓	✓
LLMTime [26]	NeurIPS '23	TS + Text	GPT-3 / GPT-4 / LLaMA	Text Digit Tokenization	Direct Prompt Likelihood	✓	✓
LLM4TS [22]	arXiv '23	TS + Text	GPT-2	2-Stage Reprogramming	Pre-train + Linear Probe	✓	✓
AutoTimes [19]	NeurIPS '24	TS + Text	LLaMA-2 / OPT	Autoregressive Patching	Next-Token Causal LM	✓	✓
CALF [23]	arXiv '24	TS + Text	LLaMA-7B	Dual-Branch Alignment	Consistency + Matching	✓	✓
S ² IP-LLM [25]	arXiv '24	TS + Text	GPT-2	Semantic Anchor Retrieval	Anchor-Guided MSE	✓	✓
UniTS [24]	NeurIPS '24	TS + Text	Unified Transformer	Task Prompt Tokenization	Multi-Task Masked Pretrain	✓	✓
VisionTS [7]	NeurIPS '24	TS + Vision	Visual MAE (ViT-Base)	Line-Chart Rendering	Masked Image Modeling	✓	✓
Time-MMD [12]	NeurIPS '24	TS + Text	Multi-Modal Benchmark	Numerical-Textual Pairs	Multi-Domain Benchmark	N/A	✓
ChatTS [9]	VLDB '25	TS + Text	LLaMA-3	Native TS Tokenization	Time Series Evol-Instruct	✓	✓
Time-VLM [8]	ICML '25	TS+Vis+Text	Vision-Language Model	Dual Augmented Learners	Cross-Modal Fusion Loss	✓	✓
TRACE [18]	NeurIPS '25	TS + Text	Dual Contrastive Tower	Channel Identity Tokens	Dual-Level InfoNCE	✓	✓
ChatTime [11]	AAAI '25	TS + Text	Multimodal Foundation	Unified Token In/Out	Joint Bimodal Pretrain	✓	✓
TimeOmni-1 [10]	arXiv '25	TS + Text	TimeOmni Backbone	Reasoning Curriculum	SFT + Reinforcement Learning	✓	✓

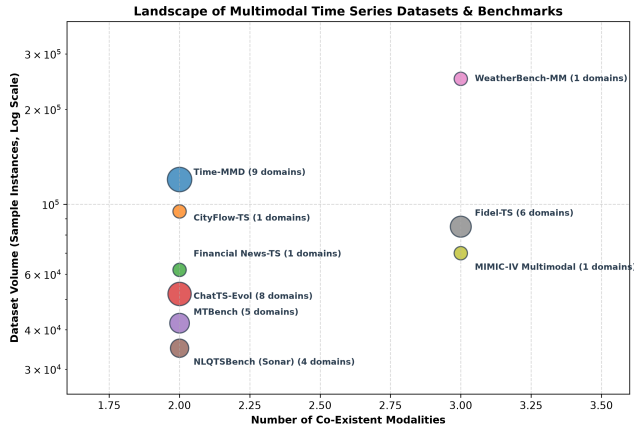


Fig. 4. Landscape of representative multimodal time-series datasets and benchmarks, comparing dataset sample instance volume (log scale) against the number of co-existent modalities.

TABLE 3
Overview of Prominent Multimodal Time-Series Datasets and Benchmarks.

Dataset / Suite	Primary Domain	Modalities	Sample Volume	Target Tasks
Time-MMD [12]	9 Diverse Domains	TS + Text	$\approx 1.2 \times 10^5$	Forecasting
Fidel-TS [13]	Multi-Domain (6)	TS + Text + Vis	$\approx 8.5 \times 10^4$	Robust Forecasting
MTBench [14]	Multi-Domain (5)	TS + Text	$\approx 4.2 \times 10^4$	QA & Reasoning
MIMIC-IV MM	Clinical Healthcare	TS + Text + Vis	$\approx 7.0 \times 10^4$	Mortality / Diag.
WeatherBench-MM	Meteorology	TS + Text + Vis	$\approx 2.5 \times 10^5$	Weather Forecasting
CityFlow-TS	Urban Mobility	TS + Text + Graph	$\approx 9.5 \times 10^4$	Traffic Forecasting

that text-augmented forecasters reduce Mean Squared Error (MSE) by 15% to 40% compared to unimodal baselines.

- Fidel-TS [13]: A high-fidelity benchmark introduced to assess model robustness under noisy, missing, or asynchronously sampled multimodal conditions across 6 domains.
- MTBench [14]: Designed specifically for evaluating temporal reasoning and Question Answering (QA) capabilities of TS-MLLMs, providing standardized metrics for causal inference, pattern recognition, and counterfactual deduction.

5.2 Cross-Task Distribution Analysis

Figure 3 cross-tabulates the distribution of published works across modality combinations and downstream tasks. Several critical patterns emerge:

- 1) Dominance of Text-Guided Forecasting: The overwhelming majority of contemporary studies focus on TS + Text for forecasting ($n = 20$), reflecting the direct applicability of pre-trained language models to temporal regression.
- 2) Emerging Frontier in QA and Reasoning: While forecasting has reached high maturity, interactive QA, semantic reasoning, and explainable anomaly detection ($n = 5$) represent the fastest-growing frontier.
- 3) Under-Explored Cross-Modal Retrieval: Despite its industrial utility for database search, cross-modal retrieval (TS $\leftrightarrow$ Text) remains nascent ($n = 1$, pioneered by TRACE [18]).

6 Open Challenges and Future Trajectories

Despite rapid progress, multimodal time series intelligence faces foundational theoretical and engineering hurdles. In this section, we delineate five open challenges and outline promising future research directions.

6.1 1. The Continuous-to-Discrete Modality Gap

While natural language and visual objects exhibit discrete, hierarchical compositional semantics, physical time series are continuous, non-stationary, and governed by fine-grained numerical gradients. Discretizing numerical time series into discrete vocabulary tokens (e.g., via scalar binning or k-means codebooks) often results in loss of numerical precision or boundary artifacts. Conversely, continuous patch projections frequently suffer from representational collapse when aligned with high-dimensional language embeddings. Developing principled cross-modal mapping functions that preserve continuous physical conservation laws while tapping into discrete semantic reasoning remains an unresolved challenge.

6.2 2. Asynchronous Sampling and Irregular Multimodal Streams

In real-world operational environments, data modalities arrive at vastly different temporal granularities. For instance, high-

frequency IoT sensor telemetry is collected millisecond-by-millisecond, financial market prices tick second-by-second, news reports appear intermittently, and satellite passes occur weekly. Current models predominantly rely on artificial resynchronization (e.g., forward filling or downsampling), which discards high-frequency signal content or introduces artificial latency. Future architectures must incorporate continuous-time neural ordinary differential equations (Neural ODEs) or state-space models (SSMs) to natively handle asynchronous, irregularly sampled multimodal streams.

6.3 3. Evaluation Integrity: Benchmark Contamination and Data Leakage

A persistent concern in the evaluation of LLM-adapted time series forecasters is the risk of training data contamination. Pre-trained web-scale language models (e.g., GPT-4, LLaMA) have ingested massive portions of GitHub repositories, academic papers, and publicly hosted datasets (such as the standard ETT, Weather, and Electricity benchmarks). Consequently, models claiming "zero-shot" forecasting prowess may in fact be performing memorized retrieval of benchmark statistical properties. Establishing contamination-resistant benchmarks, strictly partitioned temporal splits, and dynamic online evaluation protocols is imperative for scientific credibility.

6.4 4. Computational Latency and Edge Deployment Bottlenecks

Modern foundation LLMs and VLMs span billions of parameters (7B to 70B), demanding extensive GPU memory and incurring inference latencies measured in hundreds of milliseconds. In contrast, mission-critical time series applications—such as electrical grid fault isolation, surgical patient monitoring, and autonomous vehicle control—mandate sub-10-millisecond response times. Parameter-efficient distillation, specialized lightweight time-series mixers (such as Tiny Time Mixers [4]), and edge-optimized hardware accelerators represent crucial avenues for practical industrial adoption.

6.5 5. Neuro-Symbolic Agentic Frameworks and Database NLQ

Looking forward, the frontier of multimodal time series is shifting from passive predictive models toward active, autonomous analytical agents. As demonstrated by recent neuro-symbolic querying frameworks, future systems will seamlessly couple LLM reasoning controllers with formal database query execution engines (e.g., SQL and vector indexing over time-series databases). These agents will formulate hypotheses, write verification code, interrogate terabyte-scale sensor repositories via natural language, and deliver auditable, explainable insights to domain experts.

7 Conclusion

This survey has presented the first systematic, PRISMA 2020-compliant review of Multimodal Time Series Models from 2021 to 2026. By proposing a unified multi-dimensional taxonomy, providing deep mathematical dissections of cross-modal reprogramming, visual rendering, and conversational reasoning architectures, and mapping the evolving dataset

landscape, we establish a structured foundation for this rapidly expanding field. As multimodal foundation models continue to bridge the divide between continuous physical observations and discrete human reasoning, multimodal time series intelligence promises to redefine scientific discovery, industrial automation, and healthcare monitoring.

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